

UNIT-3 ASSIGNMENT

HMAC with SHA-256

98/100

a) what is the purpose of HMAC over plain hash functions?

HMAC (Hash-Based Message Authentication Code) is used to provide message authentication and data integrity using a secret key and a hash function.

Plain Hash Function

- A hash function only verifies whether data has been modified.
- Anyone can calculate the hash because no secret key is required.
- It cannot verify the identity of the sender.

HMAC

- uses a secret key along with the hash function
- Ensures:
 - Integrity: Message has not been altered.
 - Authentication: Message comes from someone who knows the secret key.
- An attacker cannot generate a valid HMAC without knowing the key.

Therefore, HMAC is more secure than a plain hash for authenticating messages.

b) HMAC Calculation steps

Given:

- Message = "HELLO"
- Key = "SECRET"
- Hash function = SHA-256

Step 1: prepare the key

SHA-256 block size = 64 bytes

Key = "SECRET" = 6 bytes

Since the ~~key~~ is shorter than 64 bytes:

- pad it with zeros until it becomes

64 bytes long.

$K = \text{SECRET}00000000000000000000\dots$

Step 2: Create Inner and outer padding

Two constants are used:

- $\text{ipad} = 0x36$ repeated 64 times
- $\text{opad} = 0x5c$ repeated 64 times

Compute:

$$K \oplus \text{ipad}$$

Where $\oplus$ represents XOR operation.

Step 3:- Calculate Inner hash

Append the message to the inner padded key:

$$(K \oplus \text{ipad}) \parallel \text{"HELLO"}$$

compute SHA-256:

$$\text{Inner Hash} = \text{SHA256}((K \oplus \text{ipad}) \parallel \text{HELLO})$$

Step 4: Calculate outer hash

$$K \oplus \text{opad} \parallel \text{Inner Hash}$$

Compute SHA 256 again:

$$\text{HMAC} = \text{SHA}_{256}((K \oplus \text{opad}) \parallel \text{Inner Hash})$$

The resulting 256-bit value is the

HMAC.

Formula:

$$\begin{aligned} \text{HMAC}(K, M) = & \text{SHA}((K \oplus \text{pad}) \\ & \parallel \text{SHA}_{256}((K \oplus \text{ipad}) \\ & \parallel M)) \end{aligned}$$

Ex:-

$$\text{HMAC}(\text{SECRET}, \text{HELLO}) = \text{SHA}_{256}$$

$$((\text{SECRET} \oplus \text{opad}) \parallel \text{SHA}_{256}$$

$$((\text{SECRET} \oplus \text{ipad}) \parallel \text{HELLO}))$$

Q) Why is MD5 not advisable in HMAC for security-critical applications?

MD5 is considered cryptographically weak due to known vulnerabilities.

Reasons:

1. Collision Attack

- Two different messages can produce the same MD5 hash!
- This weakens security.

2. Outdated Algorithm

- MD5 was designed in 1991.

- stronger algorithms such as SHA-256 and SHA-3 are now available.

3. Regulatory Restrictions

- Many security standards no longer recommend MD5.

4. Lower Security Margin

- MD5 produces only a 128-bit hash.
- SHA-256 produces a 256-bit hash, making attacks much harder.

Conclusion

For security-critical systems, HMAC-SHA-256 or HMAC-SHA-3 should be used instead of HMAC-MD5.

d) Difference Between MAC and Digital Signature

Feature	MAC (HMAC)	Digital Signature
Key used	Shared Secret Key	public / private key pair
Authentication	Yes	Yes
Integrity	Yes	Yes
Non-Repudiation	No	Yes
Verification	only sender and receiver	Anyone with public key
Speed	Faster	Slower
Typical use	Network protocols, APIs, VPNs	E-document, e-commerce certificates

MAC (Message Authentication code)

- Sender and receiver share the same secret key.
- Both can generate the MAC.
- Receiver cannot prove to a third party who created the MAC.

Digital Signature

- Sender signs using a private key.
- Anyone can verify using the public key.
- provides non-repudiation, meaning the sender cannot deny sending the message.